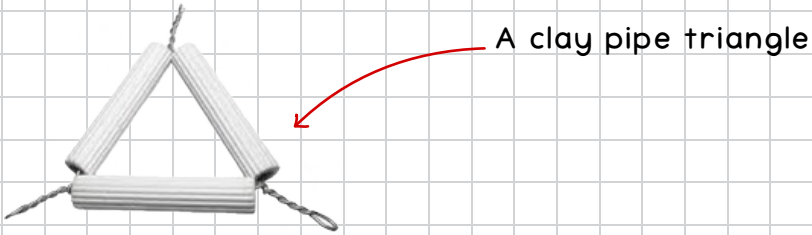
Chemistry
P5 Notes

Heating hydrated crystals to obtain anhydrous compound

A hot crucible is always allowed to cool down before weighing. This is because placing the hot crucible on the balance might damage it. Furthermore, water vapour in the air might condense on parts of its surface which are cool which might lead to inaccuracies in measuring.

The apparatus used to heat the crucible (in any case) is a tripod stand and a clay-pipe triangle.



The crucible is always heated gently at first. This is because there is water inside the crystals. If we heat rapidly at first, it might cause a rapid expansion of water which might cause the crystal to split. Split means they might break apart and ejected out of the crucible

Note: many experiments that involve heating a solid will require this apparatus too. The lid will be required or not based upon the conditions

Commenting on the validity of the experiment

Sometimes the question will ask you to comment on the validity of the experimental result, given the theoretical and experimental value and the maximum percentage error allowed. The formula is as follows,

$$\text{Percentage difference} = \left| \frac{\text{Experimental value} - \text{Theoretical value}}{\text{Theoretical value}} \right| \times 100$$

If the percentage error is within the maximum percentage error allowed, the the experiment is valid.

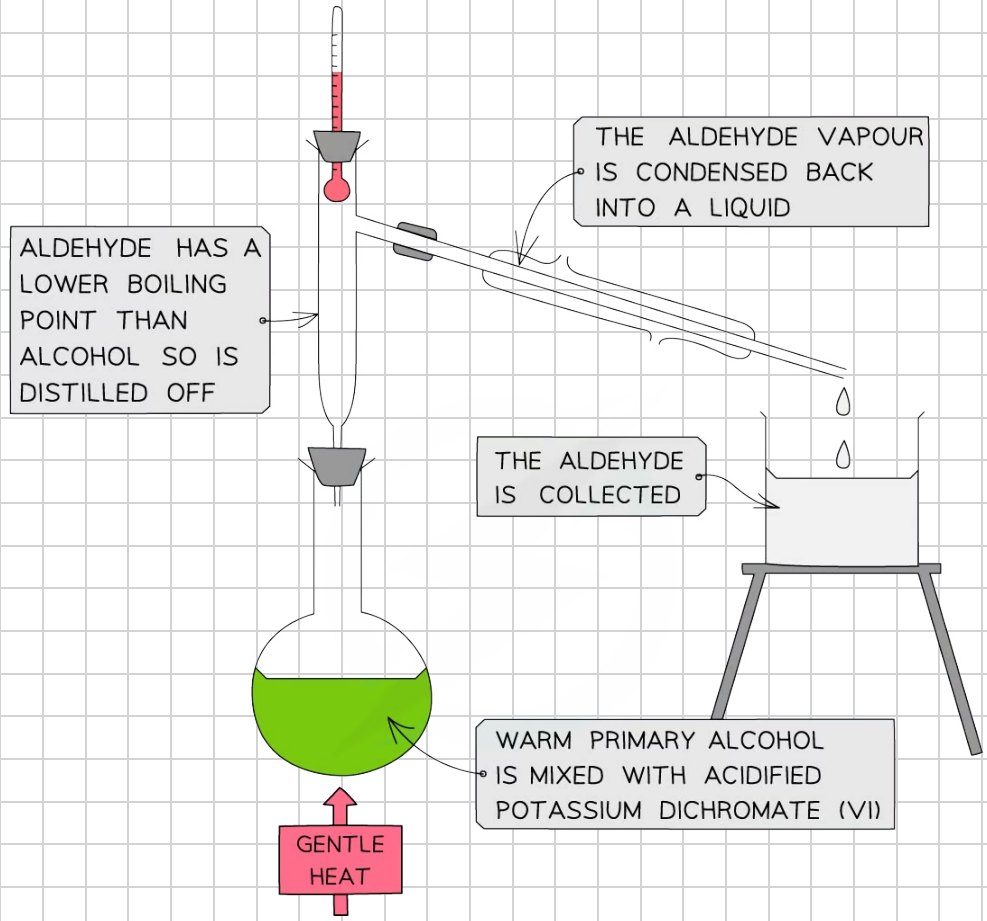
Using ice in some experiments

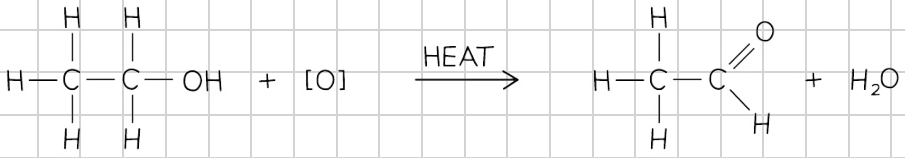
Ice is mainly used in rate experiments or when we need to do a titration. In rate experiments, we use ice to stop the rate of reaction which help then in calculating the concentration of a certain reactant through titration. They can also be used to prevent the formation of unwanted by products which typically occur at high temperatures.

How to ensure that any certain procedure is carried out at a particular temperature

Most questions like these expect you to write "thermostatically controlled water bath" but if this is not possible with the setup in the question, then you can just make sure that the surroundings are at the specific temperature and wait for the substances to equilibrate with the surrounding temperature.

Heat under distillation apparatus





ETHANOL ETHANAL

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Crystals formation

Most questions always have "how crystals are obtained" at the end. This means you not only have to explain how to make the crystals (conditions will always be given in the question e.g a substance forms crystals at lower temperature as its less soluble at lower temperatures). Crystals always have to be first filtered out (to collect them) and then rinsed with cold water to remove any impurities from the original mixture in which they were formed.

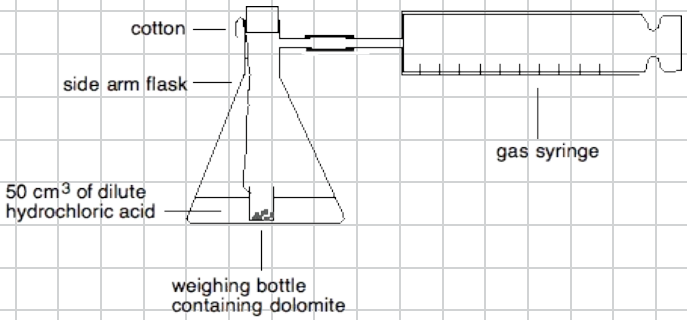
Cold water is not always used for rinsing, and the question itself will have enough information for you to decide what to use for rinsing. For example, if a substance has lower solubility at lower temperatures (which causes the crystals to form) then using cold water is recommended as the crystals will not dissolve in cold water.

It can also depend on the solubility of the impurities in the water. If the impurities are less soluble in warm water, then warm water will be used as it will be more efficient in removing the impurities

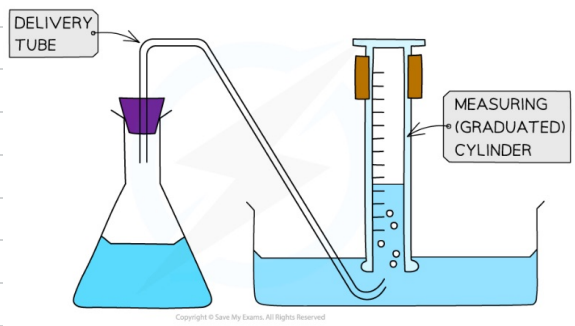
A second solvent is then used to remove the water like dry propanone

Collecting gases

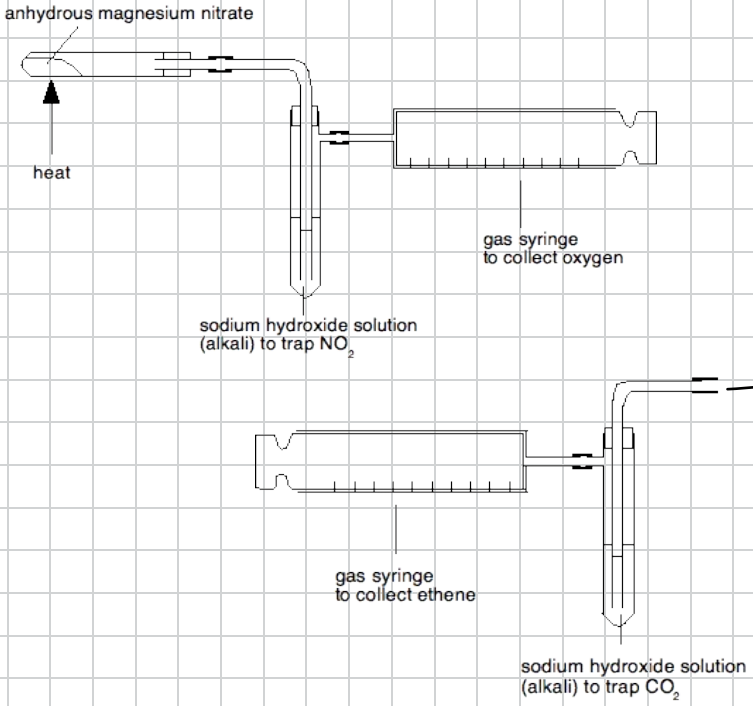
Only one gas released (method 1)



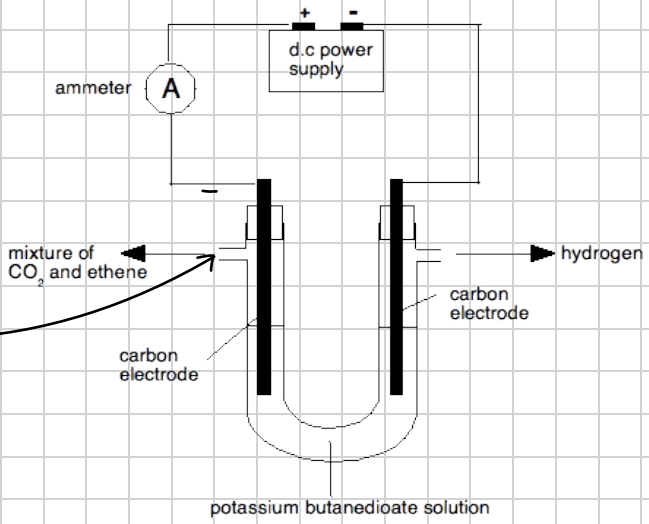
Only one gas released (method 2) (cannot be used if it reacts or dissolves in water)



2 gases released in which one gas can be absorbed



Electrolysis experiment in which 2 gases are released



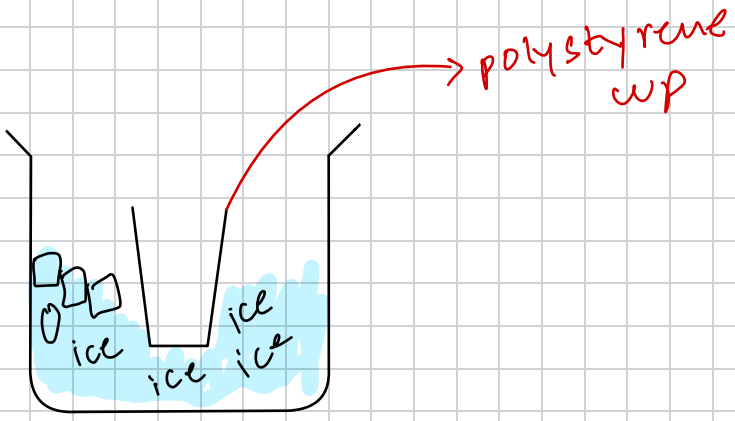
What to do before measuring gas collected?

The gas might expand due to temperature changes and also because the reaction might not be complete yet. So the 2 most important things to do is measure the gas when the volume is constant (not increasing). This is better to write as let the apparatus cool to room temperature

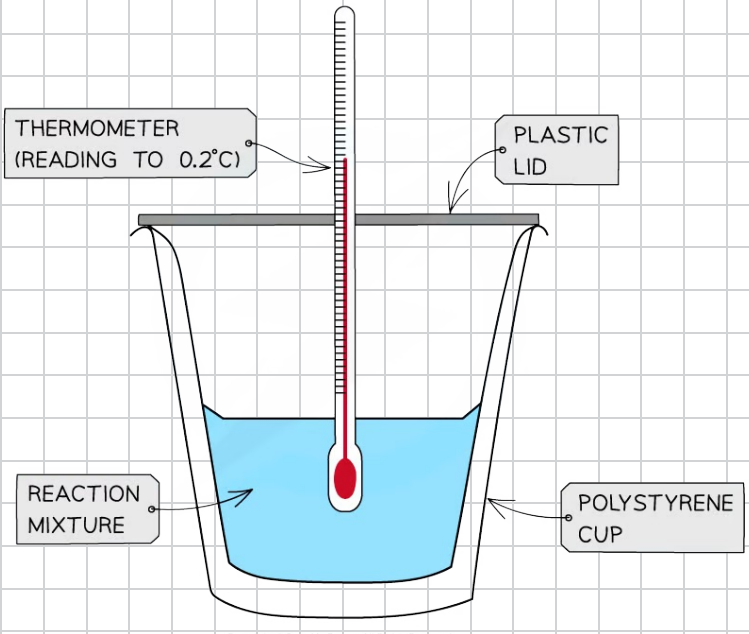
Gas	Appearance of gas	Test	Result
Ammonia (NH_3)	Colourless, pungent smell	Damp red litmus paper	Turns blue
Carbon dioxide (CO_2)	Colourless and odourless	Bubble through limewater	Limewater turns milky/cloudy
Chlorine (Cl_2)	Pale green, choking smell	Damp blue litmus paper	Turns red and quickly is bleached
Hydrogen (H_2)	Colourless and odourless	Hold a lighted splint in mouth of test tube	Burns with a 'squeaky pop' sound
Oxygen (O_2)	Colourless and odourless	Hold a glowing splint	Splint relights
Sulfur dioxide (SO_2)	Colourless, pungent chocking smell	Add to acidified aqueous potassium manganate(VII)	Turns from purple to colourless

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Apparatus to cool something



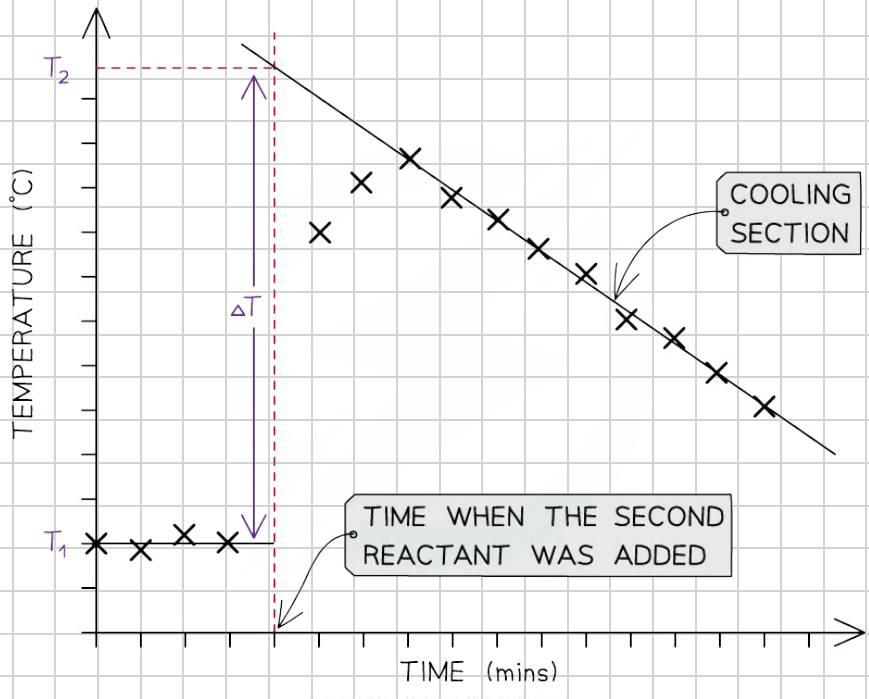
Apparatus for calculating ΔH_{rxn}



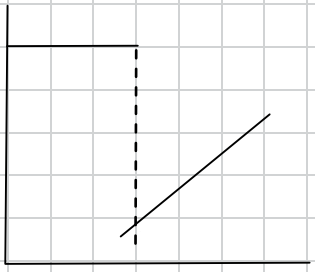
One reactant is kept in excess. Both are mixed and the temperature change is recorded,

Temperature correction graphs

- For reactions which are not instantaneous there may be a delay before the maximum temperature is reached
- During that delay the substances themselves may be losing heat to the surroundings, so that the true maximum temperature is never actually reached
- To overcome this problem we can use graphical analysis to determine the maximum enthalpy change



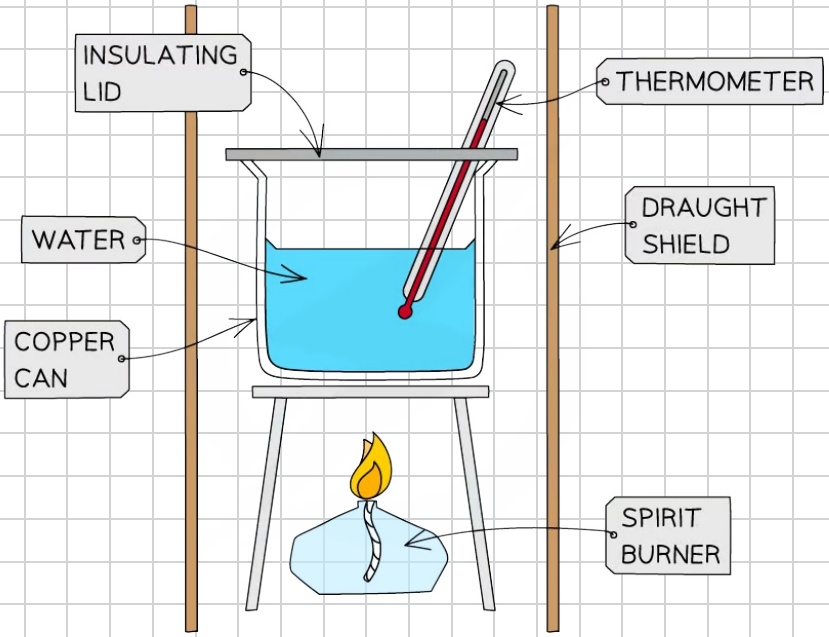
For endothermic reactions the graph will be the opposite. A straight horizontal line and then a line with a positive gradient



The steps to make a temperature correction graph are:

1. Take a temperature reading before adding the reactants for a few minutes to get a steady value
2. Add the second reactant and continue recording the temperature and time
3. Plot the graph and extrapolate the cooling part of the graph until you intersect the time at which the second reactant was added

Apparatus for calculating ΔH_{comb}



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Diluting solutions

This is a really common question that pops up. First we dissolve the solid in a beaker then transfer the solution that we need to dilute to a volumetric flask and then rinse the original container with distilled water. Then we add this to the volumetric flask. This is done to avoid leaving behind some of the solution in the original container which might have remained in the original container. Then we fill up the volumetric flask up till $250cm^3$ mark using more distilled water (This is really common question and appears in almost every past paper)

A less common type of dilution was also asked in a 2018 paper in which they wanted us to also use a $1000cm^3$ volumetric flask.

(ii) Describe how you would accurately prepare exactly $100\,cm^3$ of $5.00 \times 10^{-3}\,mol\,dm^{-3}$ urea solution, starting with 3.00 g of solid urea. Your method should use more than one volumetric flask, including one of capacity $1000\,cm^3$, and other suitable equipment.

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..... [3]

For $100cm^3$ you can also use a lesser mass of urea
but small masses produce very high percentage errors
and they told us to use 3g anyways

Make a solution of $1000cm^3$ using the 3g of urea.
This should give you a solution with a concentration of
$0.05\ moldm^{-3}$ since $M_{r_{urea}} = 60$

Using $C_1V_1 = C_2V_2$,

$$(100)(5 \times 10^{-3}) = (0.05)(V_2)$$
$$V_2 = 10cm^{-3}$$

Here we found out the volume of the old solution that we would need to dilute to get the required solution.

Now add $10cm^{-3}$ of the solution in the $1000cm^{-3}$ volumetric flask to a $100cm^{-3}$ volumetric flask
then dilute up to the mark using distilled water

Removing the oxide layer of a metal foil

This is a really common question worth 1 mark whenever they give you an experiment in which a metal foil is involved. The rate of reaction can be affected if the metal foil gets an oxide layer. So to remove it all you need to do is rub it with sandpaper and the oxide layer will be removed

Least count of burette: $0.1cm^3$

1 A saturated aqueous solution of magnesium methanoate, $Mg(HCOO)_2$, has a solubility of approximately $150\,g\,dm^{-3}$ at room temperature. Its exact solubility can be determined by titrating magnesium methanoate against aqueous potassium manganate(VII).

During the titration, the methanoate ion, $HCOO^-$, is oxidised to carbon dioxide while the manganate(VII) ion, MnO_4^-, is reduced to Mn^{2+}.

You are supplied with:
a saturated aqueous solution of $Mg(HCOO)_2$
aqueous potassium manganate(VII), $KMnO_4$, of concentration $0.0200\,mol\,dm^{-3}$

(a) (i) Write the half equations for the oxidation of $HCOO^-(aq)$ to $CO_2(g)$ and the reduction of $MnO_4^-(aq)$ to $Mn^{2+}(aq)$ in acid solution.

$HCOO^- \longrightarrow CO_2 + H^+ + 2e^-$

$MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$ [2]

(ii) Using the approximate solubility above, calculate the concentration, in $mol\,dm^{-3}$, of the saturated aqueous magnesium methanoate and the concentration of the methanoate ions present in this solution.
[A_r: H, 1.0; C, 12.0; O, 16.0; Mg, 24.3]

$Mr = 24.3 + 2(1 + 12 + 32)$
$= 114.3\,g$

$conc = \frac{150}{114.3} = 1.3123\ mol\,dm^{-3}$

$1.3123 \times 2 = 2.624\ mol\,dm^{-3}$

[2]

(iii) In order to obtain a reliable titre value, the saturated solution of magnesium methanoate needs to be diluted.

Describe how you would accurately measure a $5.0\,cm^3$ sample of saturated magnesium methanoate solution and use it to prepare a solution fifty times more dilute than the saturated solution.

Add the $5.0\,cm^3$ of the saturated solution to the volumetric flask and then fill up to the mark of $250\,cm^3$ by using distilled water.

[2]

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(iv) Before the titration is carried out, dilute sulfuric acid must be added to the magnesium methanoate.

Explain why this is necessary and also whether the volume of sulfuric acid chosen will affect the result of the titration.

The manganate ion is reduced to Mn^{2+} only in acidic conditions. The H_2SO_4 needs to be in excess as H^+ ions appear in the reactants in the overall equation to ensure that MnO_4^- is the limiting reagent which will then give an accurate titre. [2]

(v) The potassium manganate(VII) is added from a burette into the magnesium methanoate in a conical flask.

Describe what you would see when you had reached the end-point of the titration.

A pale pink permanent colour

..... [1]

(vi) 1 mol of acidified MnO_4^- ions reacts with 2.5 mol of $HCOO^-$ ions.

25.0 cm^3 of the diluted solution prepared in **(iii)** required 25.50 cm^3 of 0.0200 mol dm^{-3} potassium manganate(VII) solution to reach the end-point.

Use this information to calculate the concentration, in mol dm^{-3}, of $HCOO^-$ ions in the diluted solution.

$n(MnO_4^-) = 25.50 \times 10^{-3} \times 0.0200 = 0.00051$ mol dm^{-3}

$n(HCOO^-) = 0.00051 \times 2.5 = 0.001275$ moles.

$[HCOO^-] = \frac{0.001275}{25 \times 10^{-3}} = 0.051$ mol dm^{-3}

0.051 ..... mol dm^{-3} [1]

(vii) Use your answer to **(vi)** to calculate the concentration, in mol dm^{-3}, of the saturated solution of magnesium methanoate, $Mg(HCOO)_2$. Give your answer to **three significant figures**.

In this question they asked you the concentration in the saturated solution while the concentration you found in the previous part was for the diluted solution! It's important to read the question carefully

You would:
-find the concentration in the 5cm^3 solution using $C_1V_1 = C_2V_2$
-then divide that concentration by 2

$(C)(5) = 0.051(250)$

$C = 2.55$

$\frac{2.55}{2} = 1.275 = 1.28$ ← correct answer

0.0255 X ..... mol dm^{-3} [1]

Test for Cations

Cations	With aqueous NaOH	With aqueous Ammonia
Aluminum (Al^{3+})	**White** soluble precipitate turns colourless in excess	**White** precipitate, insoluble in excess
Ammonium (NH_4^+)	**Ammonia gas** produced on warming	
Calcium (Ca^{2+})	**White** precipitate, insoluble in excess	Faint or no precipitate
Copper (Cu^{2+})	**Light Blue precipitate,** insoluble in excess	**Light Blue** precipitate, soluble in excess to give a **dark blue** solution
Iron(II) (Fe^{2+})	**Green** precipitate, insoluble in excess	**Green** precipitate, insoluble in excess
Iron(III) (Fe^{3+})	**Red-brown** precipitate, insoluble in excess	**Red-brown** precipitate, insoluble in excess
Zinc (Zn^{2+})	**White** precipitate, soluble and turns colourless in excess	**White** precipitate, soluble and turns colourless in excess
Chromium (Cr^{3+})	**Green** precipitate, soluble in excess	**Green** precipitate, insoluble in excess

→ Add H_2SO_4, ppt will reform.